CAMSHAFT POSITION SENSOR - 4.0L (Continued)

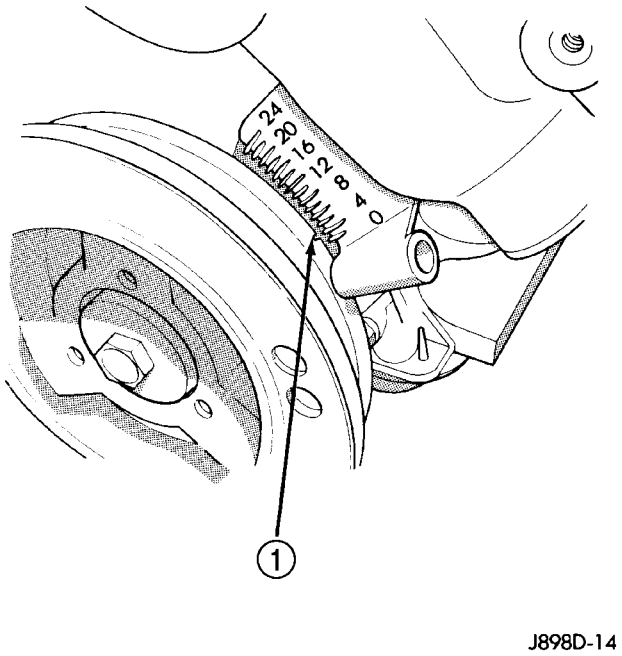

Fig. 22 Align Timing Marks - 4.0L Engine

1 - CRANKSHAFT VIBRATION DAMPER TIMING MARK

rotate engine in direction of normal rotation. Do not rotate engine backward to align timing marks.

(e) Install oil pump drive into engine while aligning into slot on oil pump. If pump drive will not drop down flush to engine block, the oil pump slot is not aligned. Remove oil pump drive and align slot in oil pump to shaft at bottom of drive. Install into engine. Rotate oil pump drive housing back to its original 4 o'clock position as viewed from right side of engine. Install hold-down clamp and bolt.

(f) Remove small screwdriver, or, factory alignment pin from housing and install plastic cover (two screws).

(7) Install CMP sensor to oil pump drive.

(8) Install sensor mounting bolt and tighten to 2 N·m (15 in. lbs.) torque.

(9) Connect CMP electrical connector to engine wiring harness.

(10) If removed, install spark plug and ignition coil rail.

(11) Connect DRB scan tool to data link connector. The data link connector is located in passenger compartment, below steering column.

(12) Gain access to "CAM/CRANK RELEARN" screen on DRB scan tool.

COIL RAIL - 4.0L

DESCRIPTION - 4.0L

A one-piece coil rail assembly containing three individual coils is used on the 4.0L 6–cylinder engine (Fig. 23). The coil rail must be replaced as one assembly. The bottom of the coil is equipped with 6 individual rubber boots (Fig. 23) to seal the 6 spark plugs to the coil. Inside each rubber boot is a spring. The spring is used for a mechanical contact between the coil and the top of the spark plug. These rubber boots and springs are a permanent part of the coil and are not serviced separately.

(1) The coil is bolted directly to the cylinder head (Fig. 24). One electrical connector (located at rear of coil) is used for all three coils.

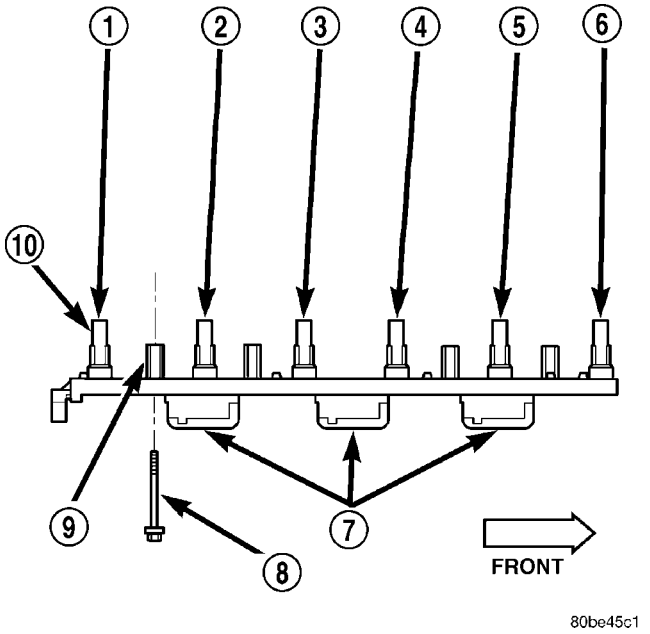

Fig. 23 Ignition Coil Assembly—4.0L 6–Cylinder Engine

1 - CYL. #6
2 - CYL. #5
3 - CYL. #4
4 - CYL. #3
5 - CYL. #2
6 - CYL. #1
7 - COILS (3)
8 - MOUNTING BOLTS (4)
9 - BOLT BASES (4)
10 - RUBBER BOOTS (6)

OPERATION - 4.0L

Although cylinder firing order is the same as 4.0L Jeep engines of previous years, spark plug firing is not. The 3 coils dual-fire the spark plugs on cylinders 1-6, 2-5 and/or 3-4. When one cylinder is being fired (on compression stroke), the spark to the opposite cylinder is being wasted (on exhaust stroke).